

Computation of a_2

The code defines a computable real number $a_2 = 2f + 1/2$, where f sums terms $(1/2)^k$ minus extremely tiny terms $2^{-k} \cdot 10^{-24}$. In the limit $n \rightarrow \infty$, the subtracted terms are **astronomically** small, so effectively $f = \sum_{k=3}^{\infty} (1/2)^k$, and hence $a_2 = 2f + 1/2$. Using the geometric-series formula (e.g. $1/2 + 1/4 + 1/8 + \dots = 1$ ¹), one finds $\sum_{k=3}^{\infty} (1/2)^k = 1 - \text{Bigl}(\frac{1}{2} + \frac{1}{4}\text{Bigr}) = \frac{1}{4}$. Thus $f \approx 1/4$ and $a_2 = 2 \cdot (1/4) + 1/2 = 1$ (up to a vanishing correction far beyond two decimal places). In other words, a_2 is essentially 1.00 to any practical precision.

Minimising total time

We now minimise the total time $T = x_1 + x_2$ subject to the dose constraint $a_1 x_1 + a_2 x_2 = 1$, with $a_1 = 1$ and $a_2 \approx 1.00$. This is a simple linear program. By LP theory, an optimal solution occurs at a vertex of the feasible line segment ². The two vertices are

- $(x_1, x_2) = (1, 0)$, giving dose $1 \cdot 1 + a_2 \cdot 0 = 1$ and total time $T = 1$,
- $(x_1, x_2) = (0, 1/a_2)$, giving dose $0 + a_2 \cdot (1/a_2) = 1$ and time $T = 1/a_2$.

Since $a_2 \approx 1.00$ (in fact slightly above 1), we have $1/a_2 < 1$. Hence the second vertex has smaller total time. Thus the minimiser is $x_1 = 0, x_2 = 1/a_2 \approx 1.00$. To two-decimal accuracy (in the ℓ^∞ norm) this gives **$x_1 \approx 0.00$, $x_2 \approx 1.00$** .

Answer: $x_1 = 0.00, x_2 = 1.00$, which uses only the faster second therapy. These values satisfy $x_1 + x_2 \approx 1$ and give the minimal total time (since $1/a_2 < 1$ when $a_2 > 1$). ¹ ²

¹ Geometric series - Wikipedia
https://en.wikipedia.org/wiki/Geometric_series

² princeton.edu
https://www.princeton.edu/~aaa/Public/Teaching/ORF363_COS323/F15/ORF363_COS323_F15_Lec11.pdf